

Quiz: Numbering systems

This quiz has 5 questions. The passing score is 4.

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Started on	Tuesday, January 14, 2025, 3:13 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 3:17 PM
Time taken	4 mins 46 secs
Grade	5.00 out of 5.00 (100%)

Question 1

Correct

1.00 points out of 1.00

The decimal number 64 is, in binary, 01000000.

Select one:

- ☒ True ✓
- ☐ False

Correct!

$$\begin{array}{cccccccc} 0 \times 2^7 & + & 1 \times 2^6 & + & 0 \times 2^5 & + & 0 \times 2^4 & + & 0 \times 2^3 & + & 0 \times 2^2 & + & 0 \times 2^1 & + & 0 \times 2^0 & = \\ 0 & + & 64 & + & 0 & + & 0 & + & 0 & + & 0 & + & 0 & + & 0 & = \\ \mathbf{64} & & & & & & & & & & & & & & \end{array}$$

The correct answer is 'True'.

Question **2**

Correct

1.00 points out of 1.00

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The decimal number 1024 is, in hexadecimal, 400.

Select one:

☒ True ✓

☐ False

Correct!

Working from hexadecimal **400** to decimal:

$$4 \times 16^2 + 0 \times 16^1 + 0 \times 16^0 =$$

$$1024 + 0 + 0 = \mathbf{1024}$$

The correct answer is 'True'.

Question **3**

Correct

1.00 points out of 1.00

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The decimal number 17 is, in binary, 00001111.

Select one:

- ☐ True
- ☒ False ✓

Correct!

$$\begin{array}{cccccccc} & 7 & & 6 & & 5 & & 4 & & 3 & & 2 & & 1 & & 0 \\ 0 \times 2^7 & + & 0 \times 2^6 & + & 0 \times 2^5 & + & 0 \times 2^4 & + & 1 \times 2^3 & + & 1 \times 2^2 & + & 1 \times 2^1 & + & 1 \times 2^0 & = \\ 0 & + & 0 & + & 0 & + & 0 & + & 8 & + & 4 & + & 2 & + & 1 \\ = & \mathbf{15, \text{ not } 17} \end{array}$$

The correct answer is 'False'.

Question **4**

Correct

1.00 points out of 1.00

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The hexadecimal number 4AB is, in binary:

Select one:

- ☐ A. 110011110000
- ☐ B. 111100001111
- ☒ C. 010010101011 ✓
- ☐ D. 011110101011
- ☐ E. 000011111100

Correct!

4 = 0100

A = 1010

B = 1011

Concatenated: 010010101011

The correct answer is: 010010101011

Question 5

Correct

1.00 points out of 1.00

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The hexadecimal number 7AAA is, in decimal, 31400.

Select one:

- ☐ True
- ☒ False ✓

Correct!

(Remember **hex A** is **10**.)

$$\begin{array}{cccc} 7 & A & A & A \\ 7 \times 16^3 & + 10 \times 16^2 & + 10 \times 16^1 & + 10 \times 16^0 = \\ 28672 & + 2560 & + 160 & + 10 = \mathbf{31402, \text{ not } 31400} \end{array}$$

The correct answer is 'False'.